

Wildfire Risk Modeling of Contra Costa County, CA

Abstract

This study models fire risk in Contra Costa County, California from 2015–2024 using a multi-variable geospatial framework integrating Landsat indices (NDVI, NBR, LST), PRISM precipitation, LandFire fuel characteristics and vegetation height, and a DEM-derived wind exposure proxy. All datasets were reprojected, normalized, weighted based on their prevalence in fire-risk literature, and combined through a 30-m weighted overlay. Historical CalFire ignitions served as a validation benchmark.

Model outputs show the highest average risk in spring and the lowest in late fall, with only modest county-wide variation across the year. Spatial maps reveal strong summer hotspots even when mean summer risk appears lower, indicating that spatial heterogeneity—rather than seasonal averages—drives fire susceptibility. Fire incidents align well with modeled high-risk zones but cluster tightly along roads, highlighting the importance of human ignition sources missing from the current model.

Sensitivity analysis shows that variables with strong spatial contrast—fuel type, NDVI, wind exposure, and vegetation height—exert the greatest influence on model variability. In contrast, Land Surface Temperature and dryness each shift outputs by <1%, reflecting relatively narrow thermal and precipitation gradients across the county.

Introduction

Wildfire risk assessment is increasingly important in California, where hotter and drier summers have intensified fire hazard in recent decades [4, 6]. Fire behavior in this region is shaped by the interaction of multiple environmental drivers, including vegetation health, fuel structure, surface heating, precipitation, and topography [2, 5, 8]. This project aims to develop a fire risk model for Contra Costa County, integrating seven key environmental factors using a weighted overlay approach. The model leverages Landsat-derived vegetation and thermal indices, PRISM precipitation, vegetation structure, and a wind exposure proxy to create 30-meter fire risk layers [9, 10, 12]. Fire occurrence data from 2015–2024 are used to validate the model and evaluate how well risk patterns align with real ignition locations and seasonal fire activity.

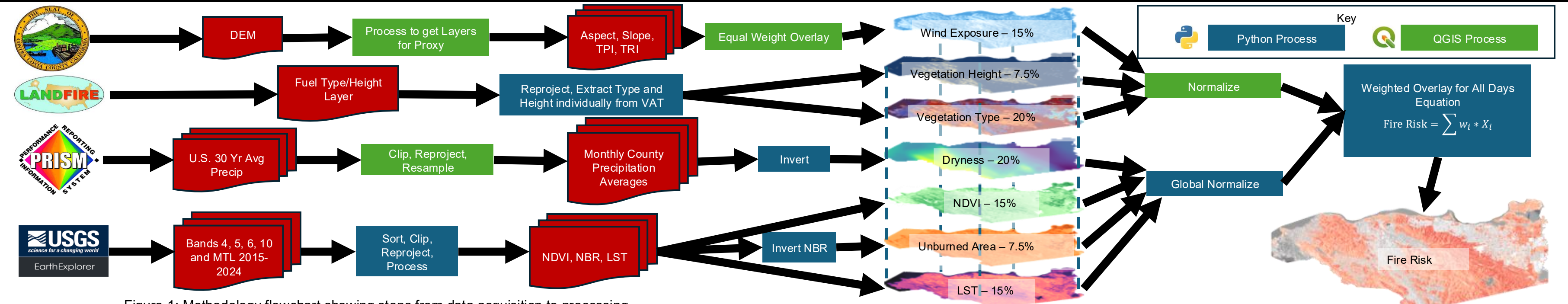
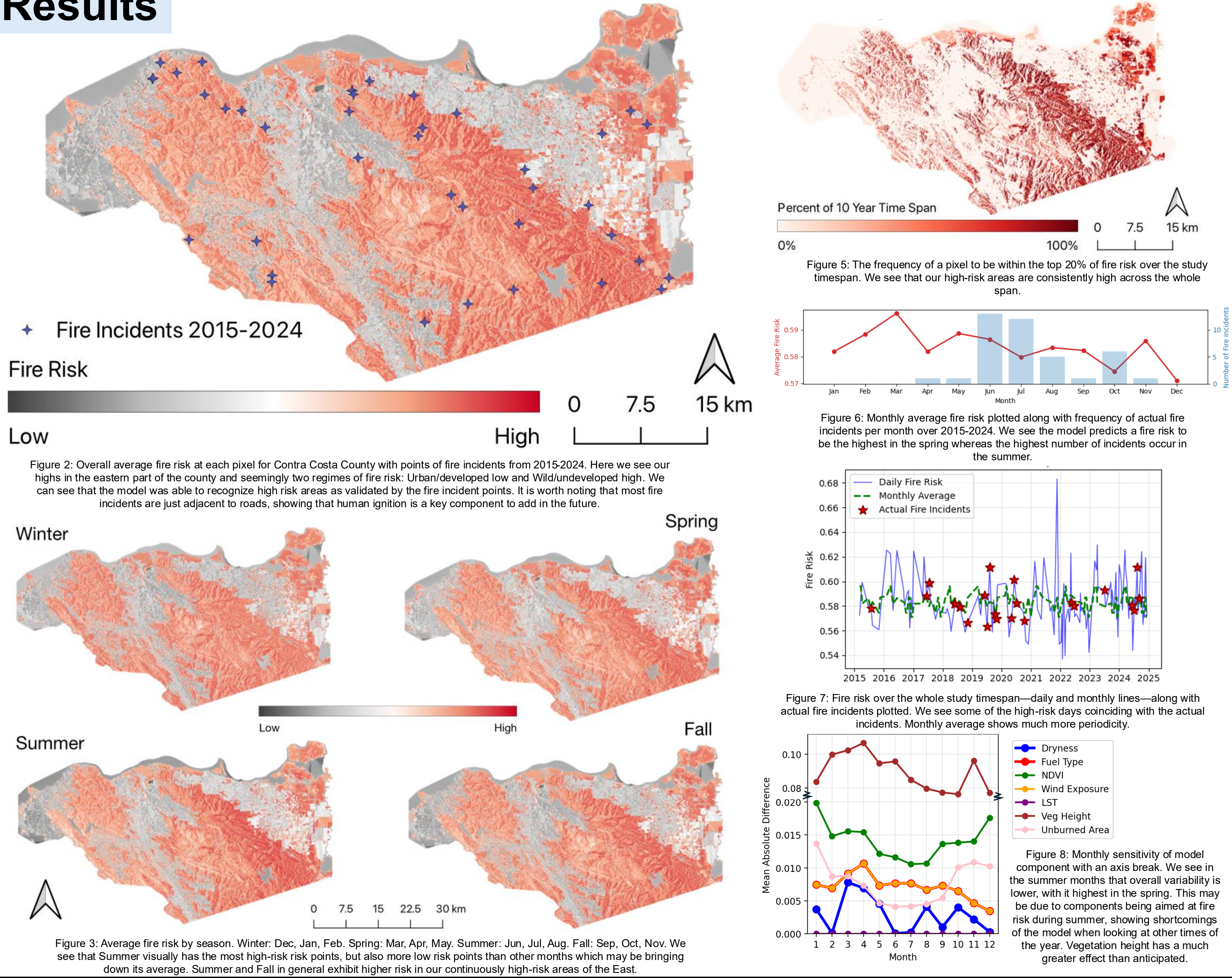
Methodology

Contra Costa County is a coastal–interior transition zone containing chaparral, oak woodlands, and extensive grasslands around the Mount Diablo foothills. All datasets were reprojected to NAD83 / UTM Zone 10N (EPSG:26910), resampled to 30 m, and clipped to county boundaries [13]. Landsat 8/9 imagery (148 scenes at ≤30% cloud cover, 1/1/2015–12/31/2024) was downloaded from EarthExplorer [16] to extract the Normalized Difference Vegetation Index (NDVI), Normalized Burn Ratio (NBR), and Land Surface Temperature (LST). Outliers caused by residual clouds or sensor noise were removed by trimming the lowest and highest 2% of pixel values. Monthly 30-year precipitation normals were gathered from PRISM [15]. Vegetation height and vegetation type were derived from the VAT file of a combined LANDFIRE layer [14], providing static fuel structure and fuel class information for the study area.

Due to limited availability of high-resolution wind data, a wind exposure proxy was derived from the county DEM [13] by combining aspect (difference with primary wind direction [17]), slope, terrain ruggedness (TRI), and topographic position (TPI) to represent wind-facing steep slopes, exposed ridges, and rugged terrain that amplify wind flow, while identifying sheltered leeward areas that reduce exposure. Fire risk was calculated using a weighted overlay of layers. Fire occurrence data (n=40) from CAL FIRE [18] were acquired to validate the model by examining spatial and temporal alignment between ignition locations and predicted risk. Processing and analysis were conducted using Python and QGIS.

Weights were assigned based on how consistently each variable appeared in the fire-risk literature. Dryness (the inverse of precipitation) and fuel type were given the highest weights (20% each) because they were the most frequently cited ignition drivers and because fires often start in dry, woody, or fine fuels where moisture deficit and fuel class strongly influence flammability [1,3,6,7,8,9,11]. NDVI was assigned 15% since it reflects vegetation stress and live-fuel moisture complementary to dryness and vegetation type [2,4,8,9,10,12]. LST (15%) and the wind exposure proxy (15%) received equal weights, as ignition temperature and wind-related terrain effects were consistently identified as major contributors to fire spread and intensity [3,5,7,10,12]. NBR (7.5%) and vegetation height (7.5%) received lower, but still meaningful, weights because they were referenced less often in the literature while still providing important information on recent burn history and vertical fuel structure [3,9,11].

Results



Discussion

Fire risk in Contra Costa County peaks in spring and is lowest in late fall, whereas most recorded ignitions occur in summer and early fall as suggested by the model (Fig. 6). This seasonal mismatch likely reflects how vegetation conditions are captured in the model with spring vegetation structure, greenness, and fuel type combinations elevating the modeled risk, even though real ignitions depend heavily on late-season drying [6] and human activity patterns not included here. County-wide averages vary only slightly (0.57–0.60), indicating that temporal fluctuations are modest compared to the strong spatial contrasts that dominate risk patterns. Summer contains the highest-intensity hotspots—particularly on the east slopes of the foothills and grasslands—despite having a lower overall mean risk than spring. This occurs because summer exhibits greater spatial spread, strongly elevated hotspots in rugged foothills but extensive low-risk coastal areas that depress the county-wide average (Fig. 3). The spatial risk maps therefore provide a more nuanced picture of fire susceptibility than simple seasonal averages. Fire incidents do fall predominantly within modeled high-risk areas and cluster almost exclusively along roads (Fig. 2). This suggests that human-caused ignitions are a major determinant of fire occurrence and a key missing component in the model [2, 12]. Integrating proximity to roads, electrical infrastructure, or recreation areas would likely improve predictive alignment. Sensitivity results show that variables with substantial spatial heterogeneity—fuel type, NDVI, wind exposure, and vegetation height—drive most model variability (Fig 8.). In contrast, LST and dryness exert <1% influence due to their relatively more narrow dynamic ranges across the region. The unexpectedly large influence of vegetation height points to the importance of vertical fuel structure, suggesting that this factor may deserve greater emphasis in future weighting schemes [11]. Overall, the model successfully captures major spatial controls on fire susceptibility and provides a strong basis for further refinement, particularly through incorporating human ignition predictors and exploring alternative vegetation metrics.

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The primary wind direction here is assumed by taking the Concord, CA Buchanan airport data which was the most complete dataset of wind direction for the county and time. The primary wind direction being a western wind. The difference of wind direction and aspect pertains to a value of 0 being a wind-facing pixel and a value of 270 being a sheltered, leeward pixel.

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